

SYSTEM ENGINEERING PART 1

Cyber-Physical Systems 2

Prof. Dr.-Ing. Michael Wiehl



Learning objectives

- Explain the concept of the system
- Input variables, function and output variables can determine
- Be able to explain a system of systems

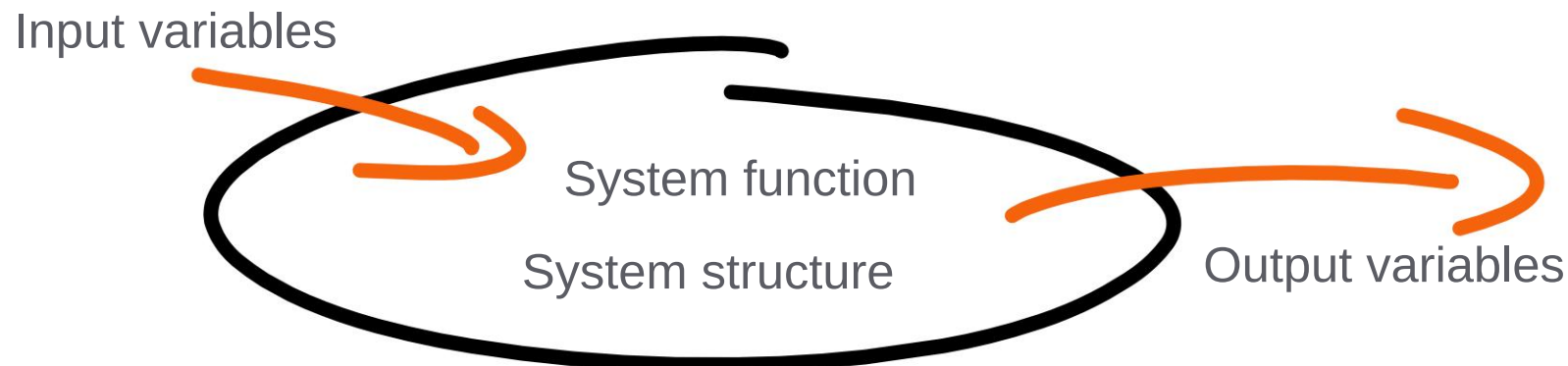


SYSTEM CONCEPT

Technical system: Definition

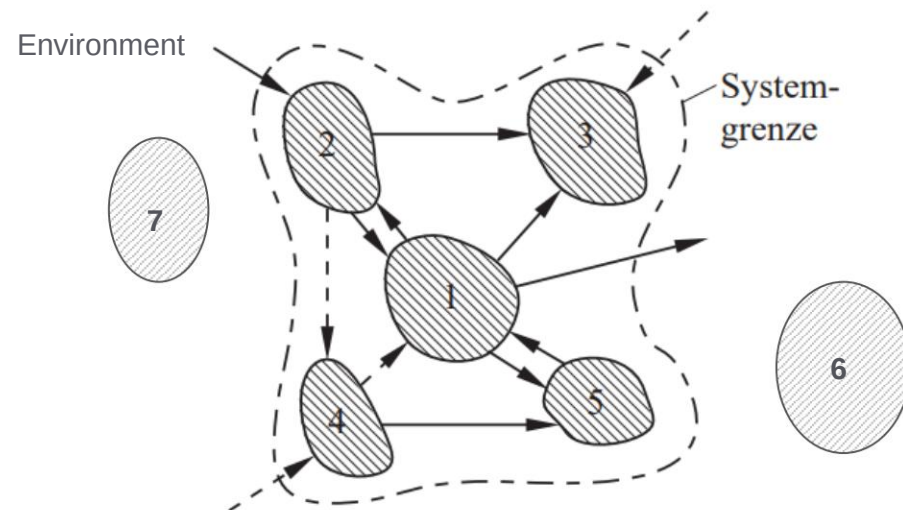
- A technical **system** is a structure that is linked by function and structure and is characterized by a **The system boundary** can be virtually separated from its environment.
- **The system function** consists of converting operational input variables into functional output variables and is supported by the structure of the system
- **System structure** consists of the entirety of system elements, their properties and Interactions

(Source: Prof. Czichos, BHT Berlin)



Technical system: Definition

- Generally speaking, **systems are defined as part of reality** . They represent a delimited arrangement of interacting entities and, because of this characteristic, have a relative nature.
- The **boundary between a system and its environment** can be described by a surface, the **system boundary** . More precisely, a system is always a collection of subsystems that **are informationally** interconnected with each other and with the environment.



(Source: Mechatronics 4th edition, Heimann, Hanser-Verlag)

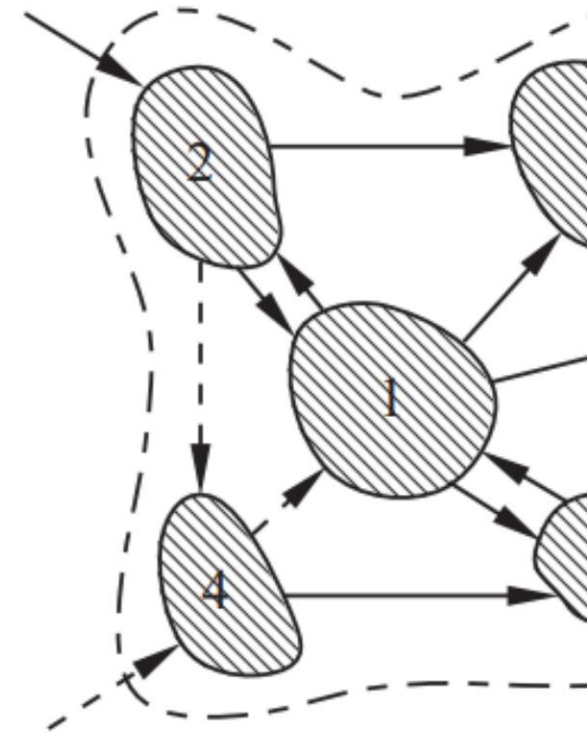
Technical system: Definition

- Generally speaking, systems are those that possess the following properties are given:
 - A system fulfills a **specific function**, i.e., it can be defined by a system purpose that the observer ascribes to it.
 - A system consists of a **specific constellation of system elements and interrelationships** that determine its functions.

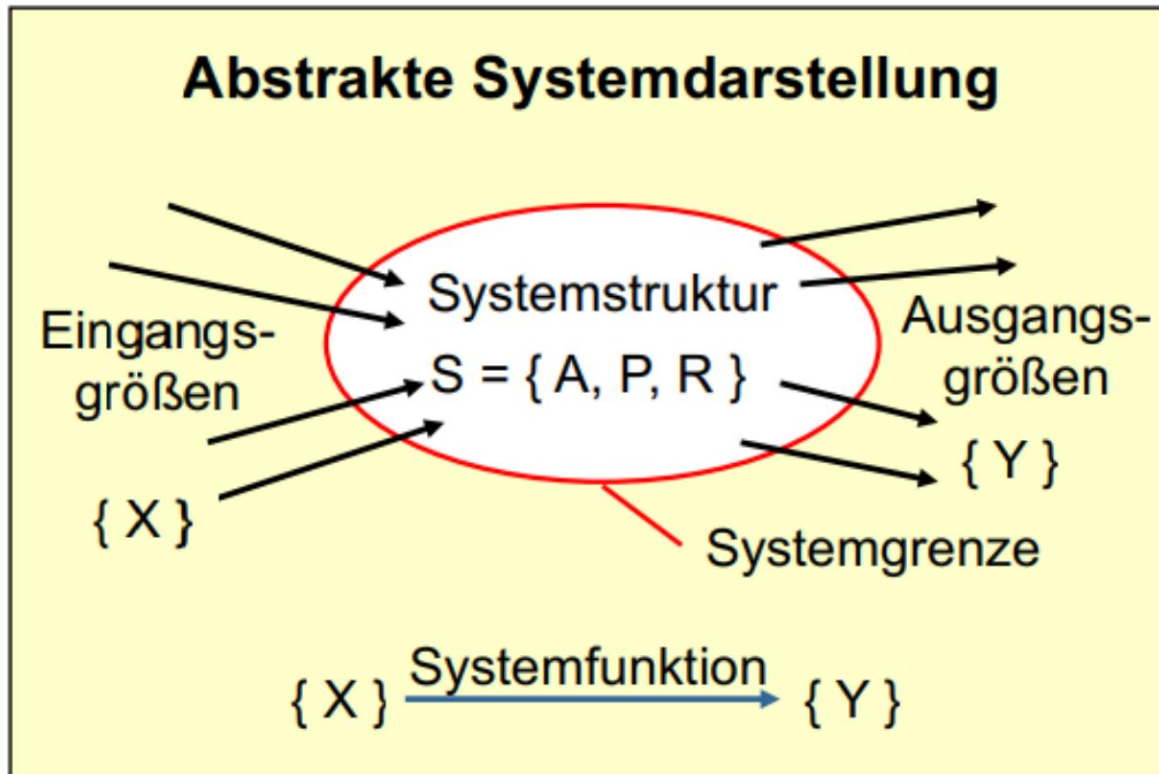
A system loses its system identity when its system integrity is destroyed.

This implies that **a system is indivisible**, i.e., there are elements and relations within this system whose removal or destruction would no longer allow the fulfillment of the original system purpose or function: The system identity would have changed or been completely destroyed.

(Source: Handbook of Robotics, 2nd edition, Matthias Haun, Springer/Vieweg)



Technical system: Definition



Prof. Dr. H. Czichos
BHT Berlin

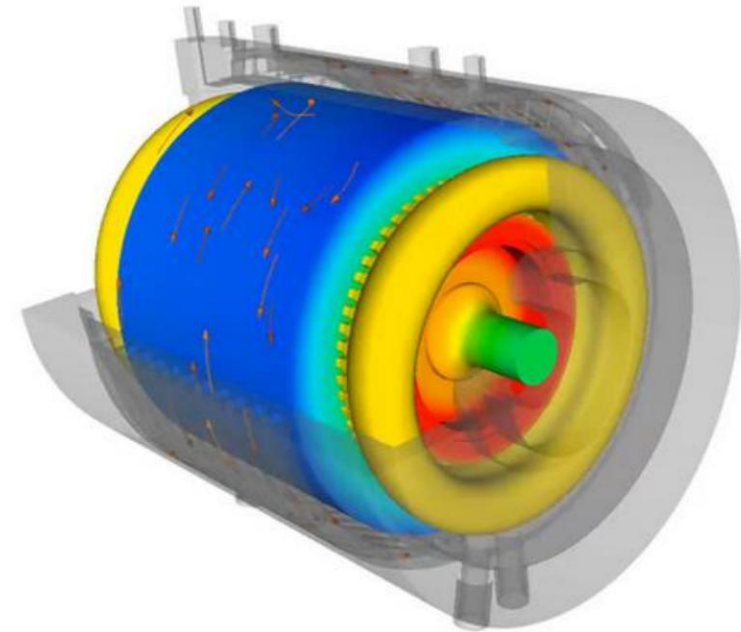
Systemstruktur: $S = \{ A, P, R \}$, mit
 A: Systemelemente
 $A = \{ a_1, a_2, \dots, a_n \}$,
 (n Anzahl der Elemente)
 P: Eigenschaften der Elemente
 $P = \{ P(a_i) \}$
 R: Wechselwirkungen der Elemente
 $R = \{ R(a_i, a_j) \}$

Systemfunktion: $\{ X \} \rightarrow \{ Y \}$
 X: Eingangsgrößen
 Y: Ausgangsgrößen

Mechatronik-Skript 2020

Technical system using an example

- Input variables: current, voltage
- Output variables: torque,
Heat output
- Systemelemente: Stator, Rotor
- Rotor characteristics: efficiency,
maximum torque
- Interaction between stator and rotor: alternating
magnetic field



(Image source: webmotor.org)

Transfer to software: In software, for example, there are...

Input variables

Interactions

Systemelement

System function

Characteristics

Output variables



SUBSYSTEMS AND SUPERSYSTEME

Define a system ...

- Assumed:

- 1 = Embedded System

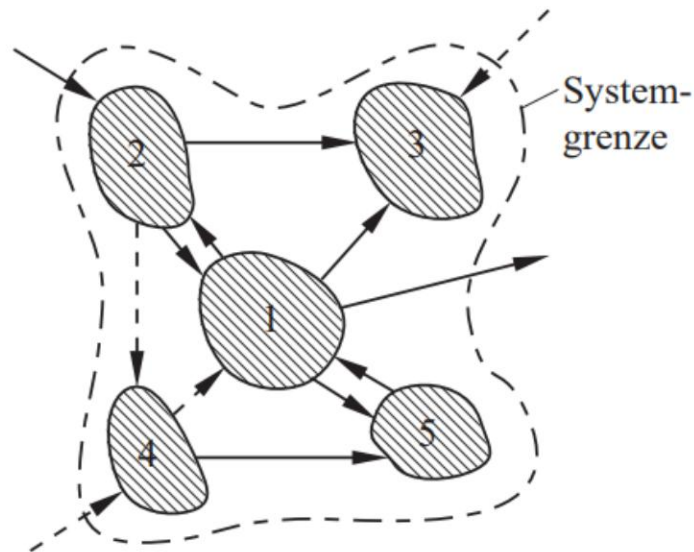
- 2 = Rule system

- 3 = KI-System

- ...

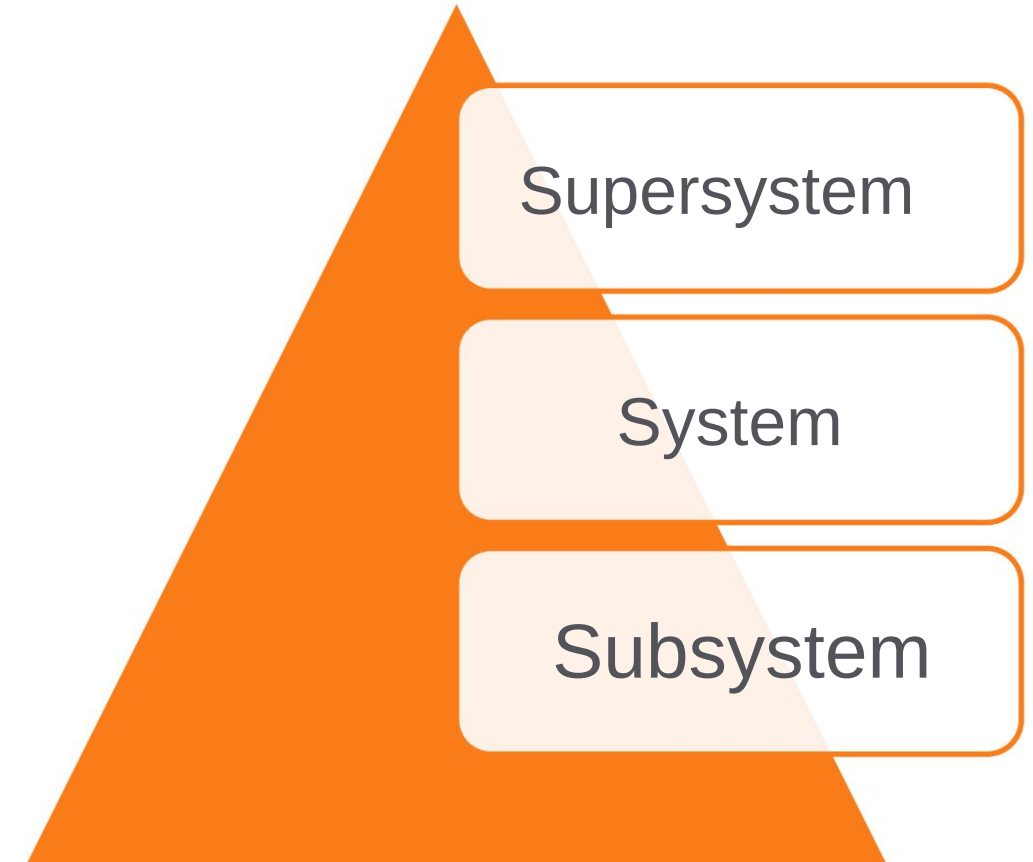
- System boundary defines new System

- Are its parts then
Elements, modules, or
components?



Supersystem and subsystem

- Depending on the reference level:
 - Combination of systems leads to supersystems
 - If parts of a system are themselves systems, they can be called subsystems (or partial systems).
to be designated
- Alternatively: instead of “Supersystem”, “System of Systems” is also used.



System of Systems

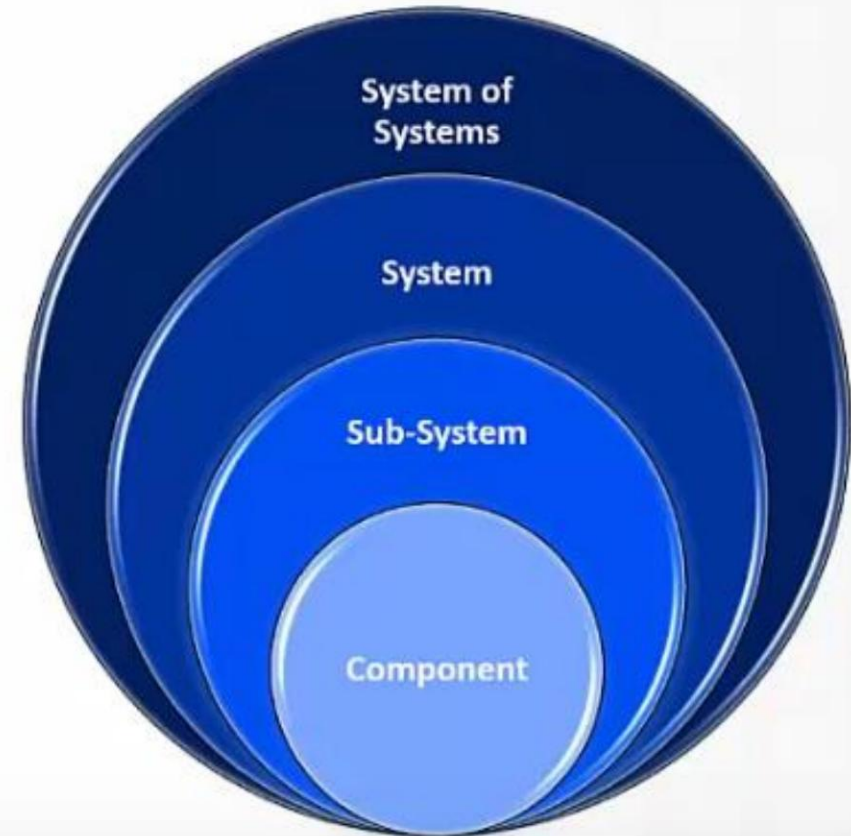
Fundamental Concepts

A System:

- Performs a function, transforming inputs to outputs
- Is a collection of interacting components with a common goal

A Subsystem:

- Can be considered a system
- Therefore, the analysis and specification of a system is hierarchical and iterative
 - System
 - Segment
 - Subsystem
 - ...



(Quelle: Vortrag „How to Engineer a System of Systems Using CORE“, VitechCorp,
<https://www.youtube.com/watch?v=5ud0uPhBjg4>)

Difference System of Systems to System

- 1. Operational independence of the elements:** The elements that make up a system of systems are themselves completely self-contained systems, meaning that these systems can **be started up** and used **independently** of one another . They can therefore also leave one system of systems to join another. This means that the **coupling between the subsystems is very loose.**

(Source: "What exactly is a System of Systems?", Stephan Roth, Training, Consulting & Coaching, roth-soft.de)

Difference System of Systems to System

1. Operational independence of the elements
2. **Independent management of the elements:** The subsystems of a SoS can not only
They work independently of each other; they are usually also **developed and procured separately**.
Their life cycle is independent of the life cycle model of the System of Systems.

(Source: "What exactly is a System of Systems?", Stephan Roth, Training,
Consulting & Coaching, roth-soft.de)

Difference System of Systems to System

1. Operational independence of the elements

2. Independent management of the elements

3. Evolutionary development: A fully developed system of systems does not exist.

Its development and existence is evolutionary, meaning that throughout its life cycle, functions – and therefore also subsystems – are added, removed, or changed based on experience.

(Source: “What exactly is a System of Systems?”, Stephan Roth, Training, Consulting & Coaching, roth-soft.de)

Difference System of Systems to System

1. Operational independence of the elements

2. Independent management of the elements

3. Evolutionary Development

4. **Emergent behavior:** Emergence (from the Latin *emergere* for “the emergence”, “the Emergence (or “rising up”) is the spontaneous emergence of new properties or functions of a system as a result of the interaction of its elements. [...] Applied to a system of systems, this means that through the **combination of subsystems**, certain properties and services for the user emerge that **can no longer be clearly assigned to specific subsystems**).

(Source: “What exactly is a System of Systems?”, Stephan Roth, Training, Consulting & Coaching, roth-soft.de)

Difference System of Systems to System

1. Operational independence of the elements
2. Independent management of the elements
3. Evolutionary Development
4. Emergent behavior
5. **Geographical Distribution:** The geographic extent and distribution of the subsystems of a system of systems is often very large. “Size” is certainly a vague and relative concept, but in relation to a system of systems, it means that its subsystems, due to their spatial distance from one another, can essentially **only** exchange information (data), and **not** significant amounts of mass, matter, or energy.

(Source: “What exactly is a System of Systems?”, Stephan Roth, Training, Consulting & Coaching, roth-soft.de)

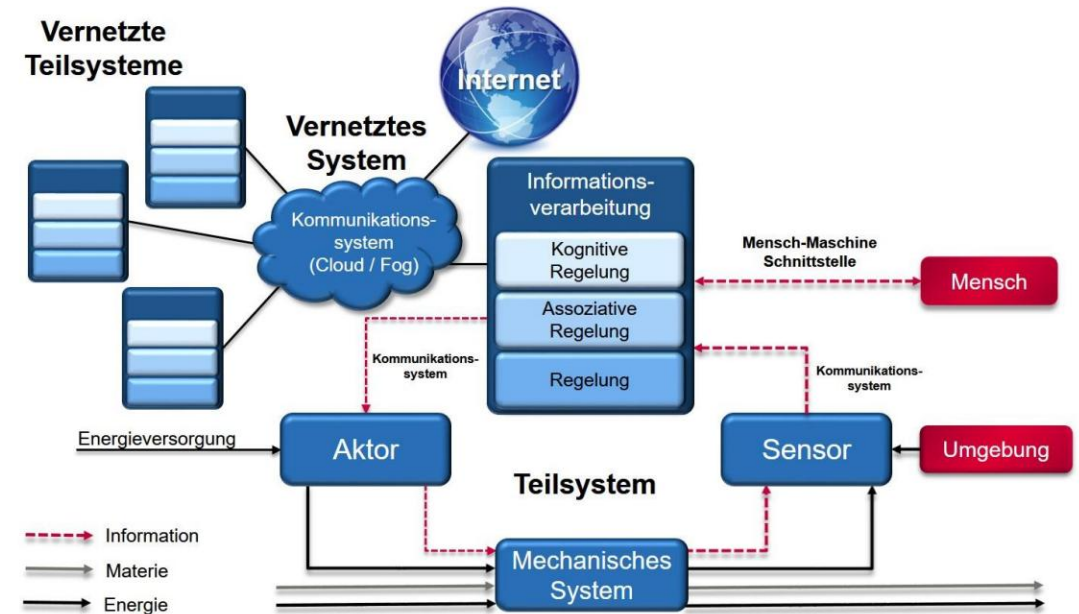
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Characteristics of a SoS using the example of Smart Factory

1. Operational independence of the elements
2. Independent management of the elements
3. Evolutionary Development
4. Emergent behavior
5. Geographical distribution



(Source: ZHAW Zurich University of Applied Sciences)

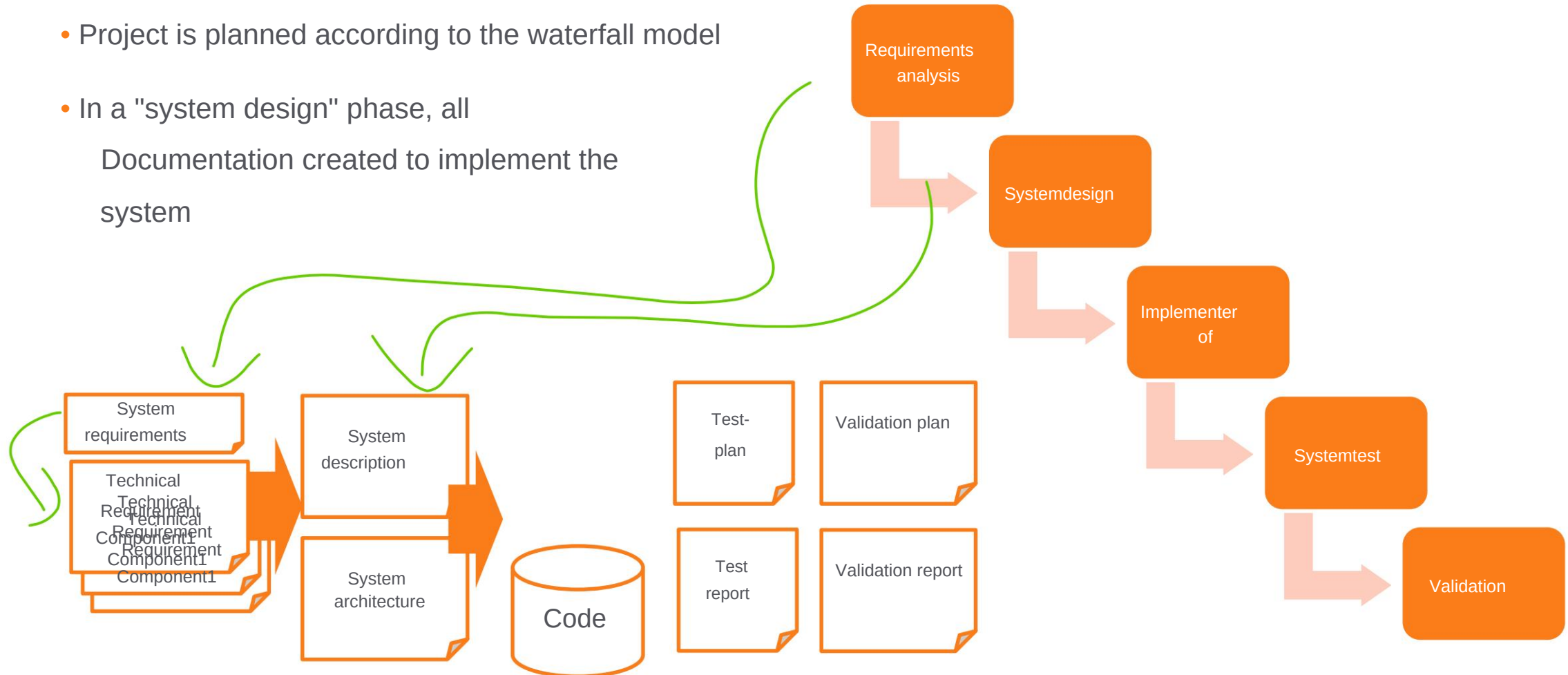
SYSTEM ARCHITECTURE CREATE

System architecture or architecture description

- The term architecture comes from the Latin "architectura" and means building art.
- It has been common practice in building construction for centuries to design architecture
- The system architect's task is to understand the requirements for a system or product and to develop a solution from them.
To design a system that can efficiently, cost-effectively, and safely meet these requirements
- **System architecture ...**
 - ... is the visualization of a system
 - ... enables review and discussion
 - ... is a guideline for development

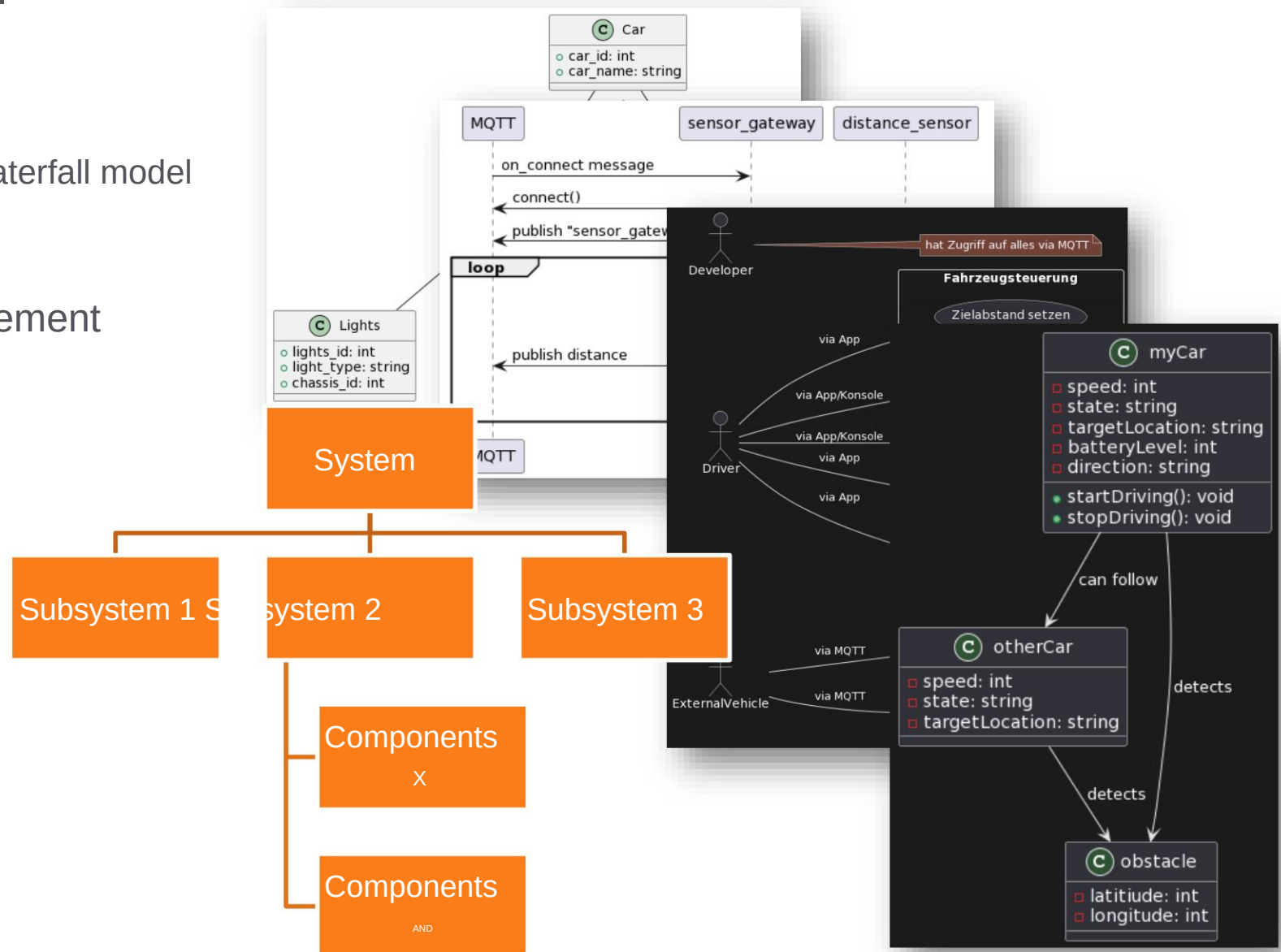
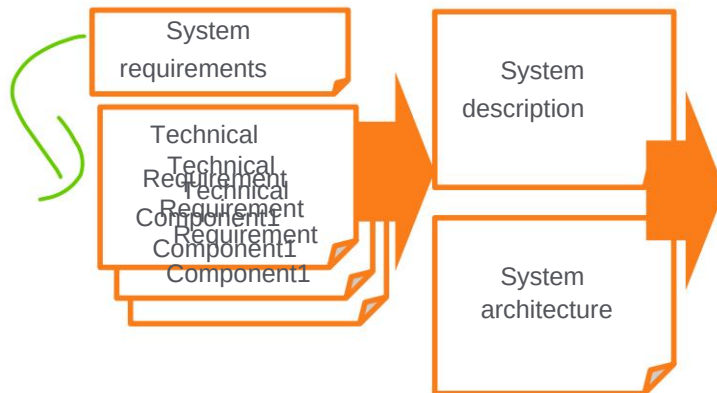
Architecture as a result of system design in the waterfall model

- Project is planned according to the waterfall model
- In a "system design" phase, all Documentation created to implement the system



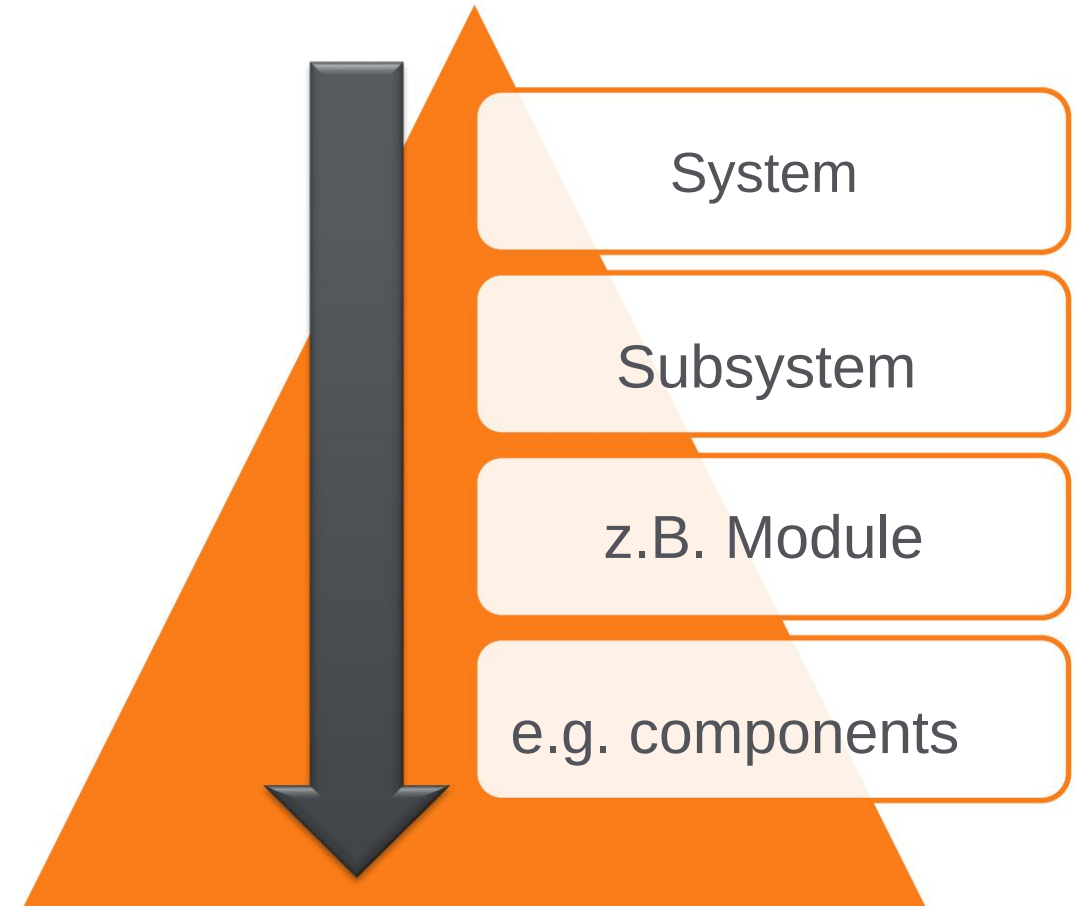
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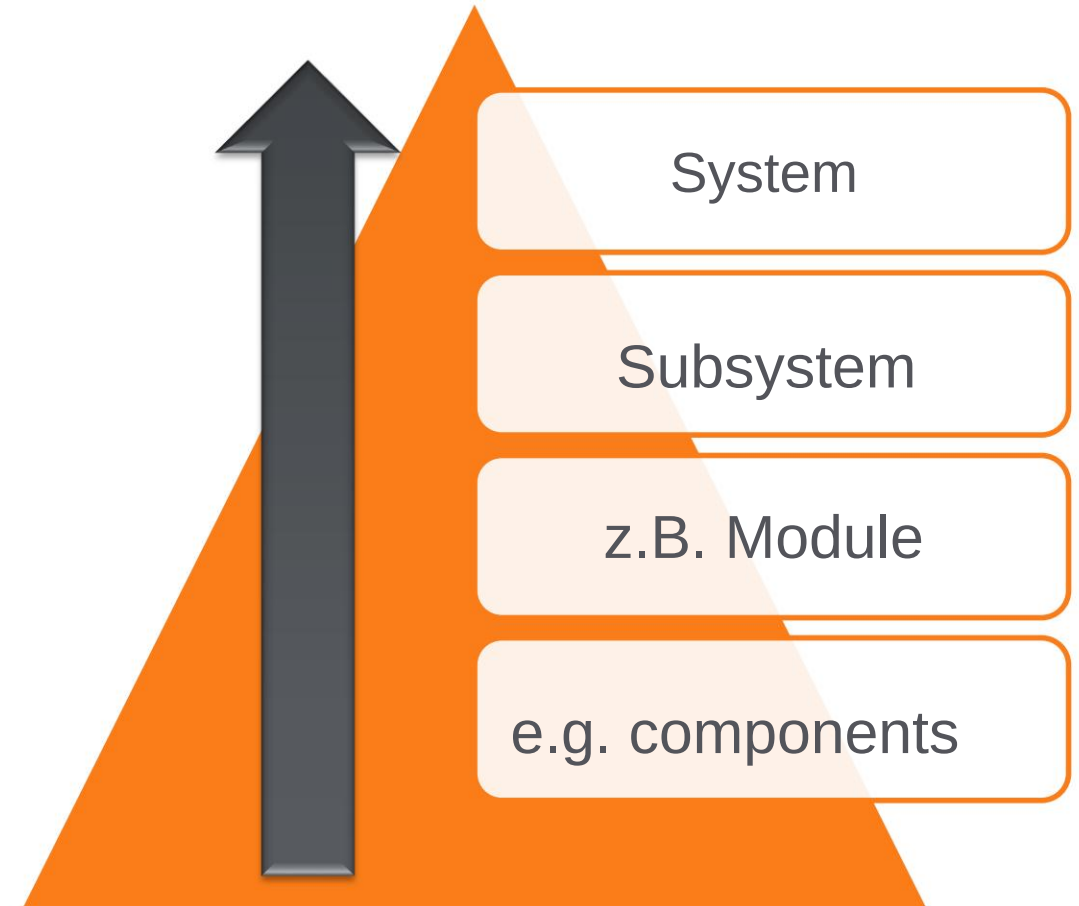
Creating the system architecture top-down

- The system is broken down into individual components
- Components are described individually and designed on paper
- Works well if some components are readily available for purchase/use.
- Can serve as a basis for proceeding according to the V-model



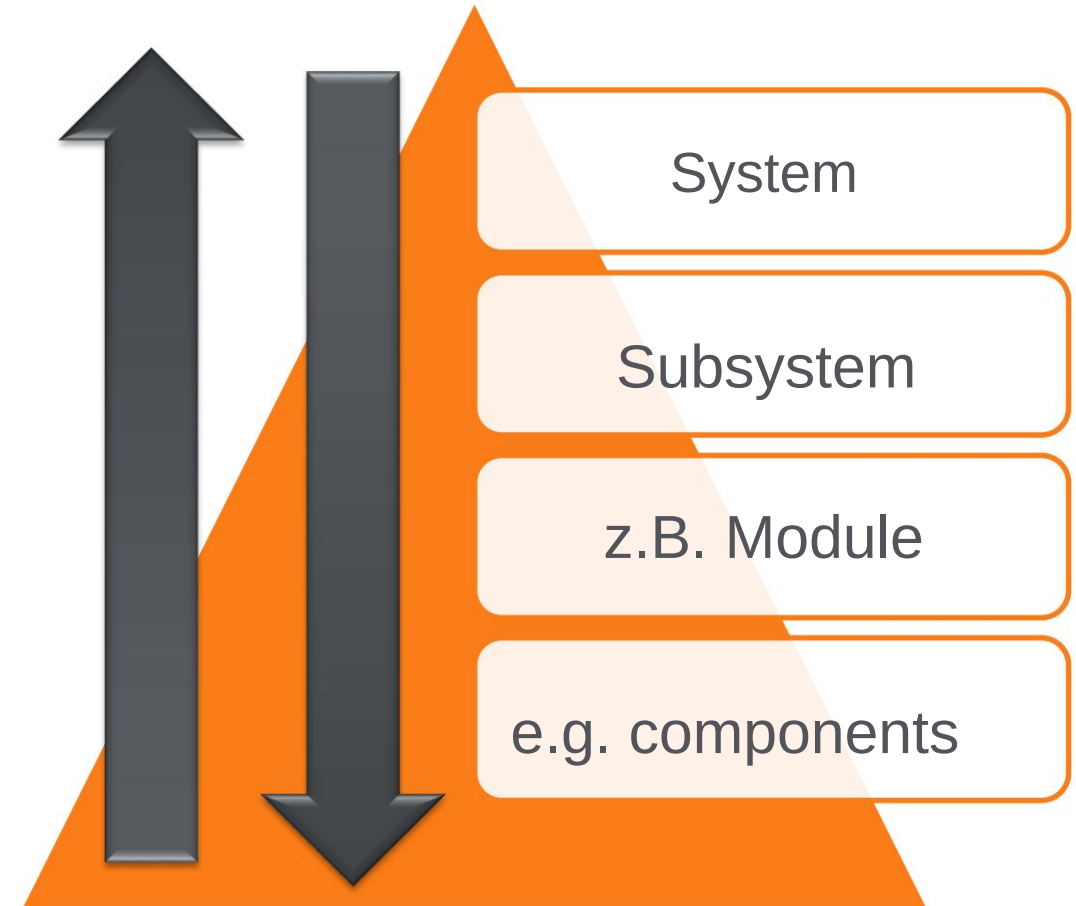
Creating the system architecture from the bottom up

- First, components are identified and described
- More complex structures and the overall system are then described.
- Components can be taken from the Design Repository come
- Can serve as a basis for proceeding according to the V-model



Creation of the hybrid system architecture

- Bottom-up and top-down approaches are combined and carried out sequentially.
- Useful in cases of high complexity or for agile processes procedure in the phase where the architecture is created becomes



Exercise on system definition

Goal: Design of a roller shutter control system for a smart home



System

Subsystems

z.B. Module

e.g. components

How would you visualize or document a software architecture?



DESIGN PROCESS

The aim of the design

- Comprehensive analysis of all relevant system requirements
- Identify and consider (physical) dependencies
- Create necessary models and use them for design
- Determine component requirements
- Select suitable components
- Verify suitability using theoretical methods and practical tests

What steps would you take in the design of a robot, from idea to implementation?



Divide and conquer

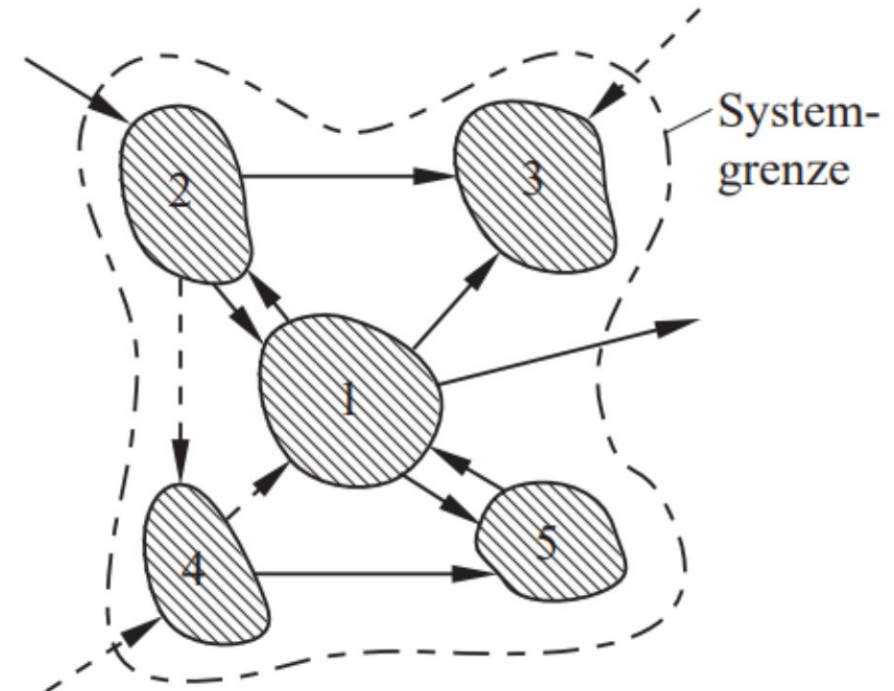
- Divide et impera (Latin for divide and rule) is an idiom (in the Latin imperative); it recommends splitting a group to be defeated or controlled (such as a people) into subgroups with conflicting interests.
- primarily used in computer science ÿ refers to a paradigm for the design of efficient Algorithms
- Can also be used for systems analysis: **Divide the system into subsystems, which are easier to analyze. are manageable**

Divide and conquer

- Method:
 - breaking the problem down into many sub-problems
 - recursive solving of the sub-problems
 - reassembling completed partial solutions into a whole
- Goal: To create an entire complex system, e.g., a cyber-physical system

Divide and conquer in systems

- Systems can be divided into subsystems or partial systems.
- Parts of the system must not interact with others overlap
- No parts may be missing

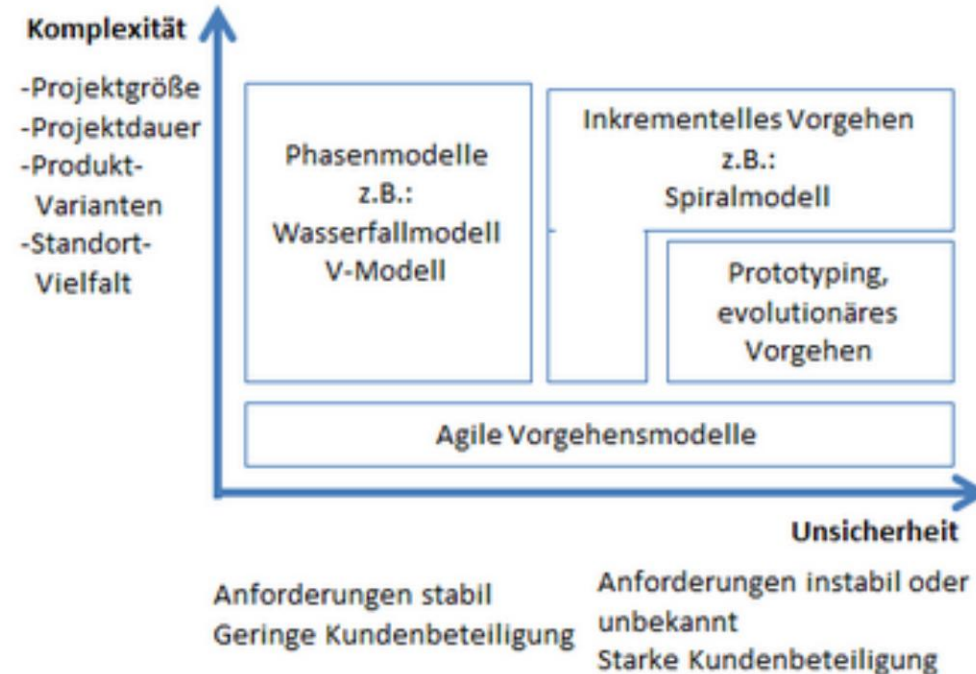


Project methodologies

- Which is important for the development of a system Suitable? – all

But ..

- Advantage of the V-model: Division into components, which are tested separately

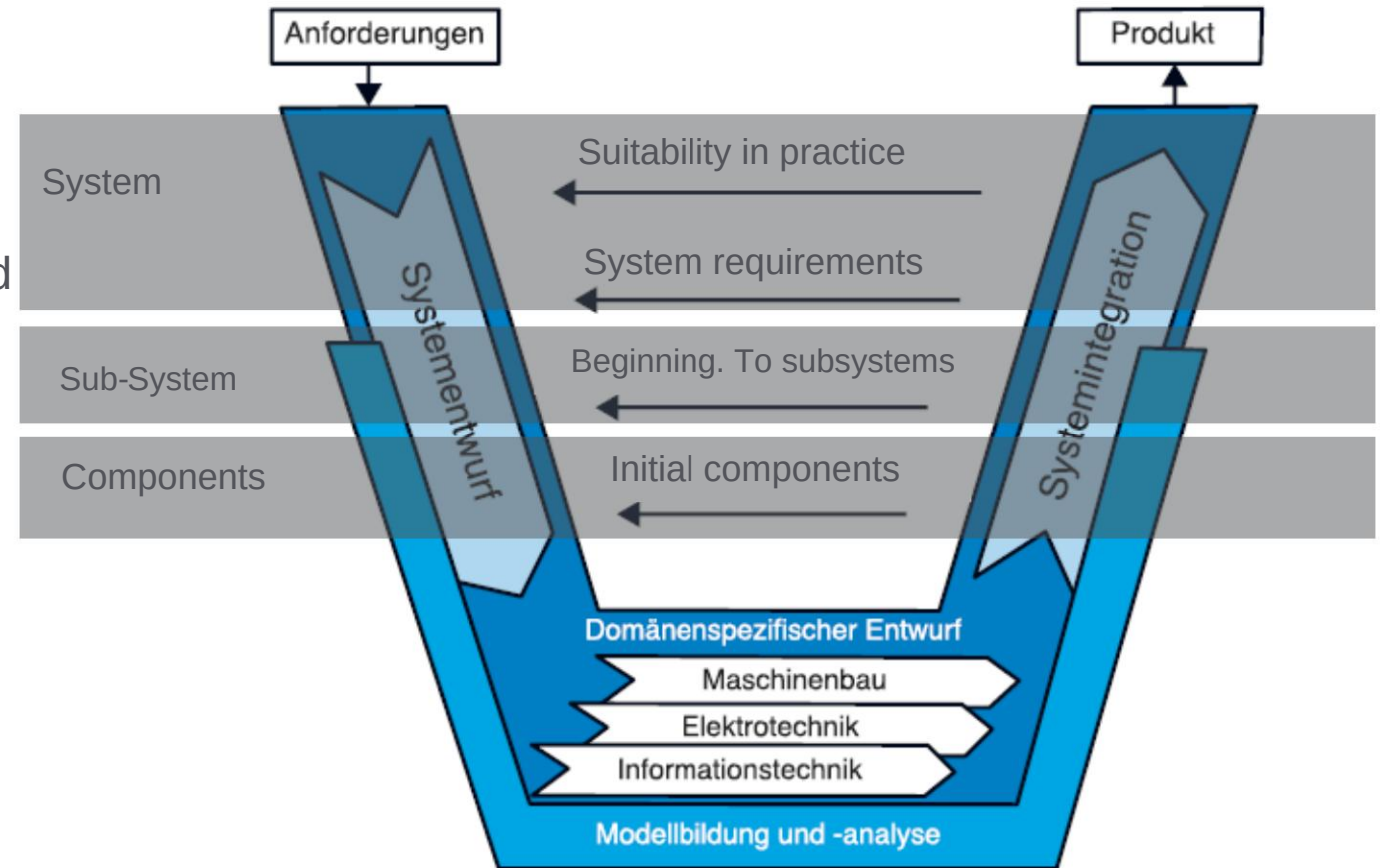


*Timinger, 2017 S. 243 in Anlehnung an
Ebert, 2014*

Ebert (2014), Systematic Requirements Engineering, dpunkt.verl.
Timinger (2017), Modern Project Management – Achieving Success with Traditional, Agile and Hybrid Approaches, Wiley

System Engineering im V-Modell

- The system is, for example, used in the design process designed top-down
- System requirements become Requirements for subsystems are derived
- During system integration, sub-Systems pre-tested

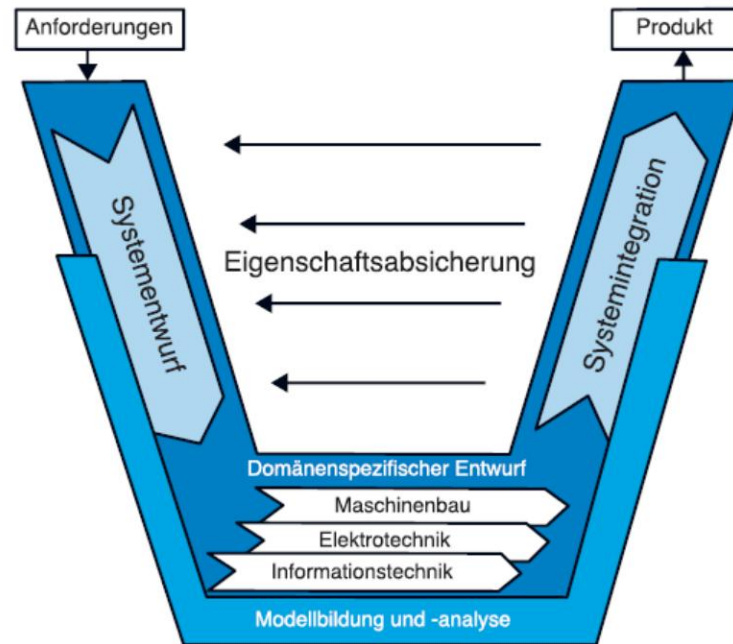


V-model according to VDI 2206

Exercise on system definition

Goal: Design of a roller shutter control system for a smart home

How many stages would you divide the process into before starting implementation? Add activities like ".. design" or ".. test" to the left/right of each stage.



Which visualization would be useful in this application.

